Toy Experiment: Setup



CLEAN Write a short message to my landlord. I'm Minji Park 0.999 from 41 Maple Street 0.969, and my heater is broken. He can call me at 010-2345-6789 1.000.

NOISY write a short message to my landlord im minji park 0.686 from 41 maple stret 0.921 and my heater is broken he can call me at 010 2345 6789 0.985

GLiNER masks all three values in both prompts, but the lowercase name drops from 0.999 to 0.686, close to the 0.5 cut-off.

Highlight = span GLiNER masked; number = its confidence. Detector: urchade/gliner_multi_pii-v1. LLM: gpt-5.5 through the OpenAI Responses API.

Toy Experiment: Example 1 (Clean)

YOU WRITE Write a short message to my landlord. I'm Minji Park from 41 Maple Street, and my heater is broken. He can call me at 010-2345-6789.

LLM RECEIVES Write a short message to my landlord. I'm <NAME_1> from <ADDRESS_1>, and my heater is broken. He can call me at <PHONE_1>.

LLM ANSWERS Hi, this is <NAME_1> from <ADDRESS_1>. My heater is broken and needs repair. Please call me at <PHONE_1> when you can. Thank you.

YOU READ Hi, this is Minji Park from 41 Maple Street. My heater is broken and needs repair. Please call me at 010-2345-6789 when you can. Thank you.

Answers are shown in full, as returned by gpt-5.5.

Toy Experiment: Results

	Example 1 (clean)	Example 2 (noisy)
PRIVATE?	**Yes** no real value sent	**Yes** no real value sent
ENOUGH CONTEXT?	**Yes** all 3 placeholders used	**Yes** all 3 placeholders used
USABLE ANSWER?	**Yes** all 3 values restored	**Yes** all 3 values restored

- **The loop works:** GPT-5.5 wrote the message without seeing any real value, and every placeholder was restored. Same result in 3 runs.
- **RQ2 hint:** typing the name in lowercase cut its confidence from 0.999 to 0.686; nothing leaked, but close to the cut-off.
- **The noise comes back:** unmasking restores what the user typed, so the noisy answer reads “this is minji park from 41 maple stret”.